Jenkins task 1

Jenkins Installation & Configuration Task

Task Objective

- Install Jenkins and run it on **port 8081**
- Secure Jenkins server
- Create limited users (**DevOps**, **Testing**)
- Configure labels and restrict jobs using labels
- Create **3 sample jobs** from a public GitHub repo
- Create **1 job** from a private GitHub repo

1. Install Jenkins and Run on Port 8081

—> launch one ec2 and follow below steps

Prerequisite to Install jenkins in java.

Step 1

Download jenkins Repo:

======================

sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat-stable/jenkins.repo

Import the jenkins key:

======================

sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key

Update ec2:

==========

sudo yum upgrade

```
[ec2-user@ip-172-31-65-24 ~]$ sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat-stable/jenkins.repo
--2025-12-23 09:05:03--  https://pkg.jenkins.io/redhat-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.38.133, 2a04:4e42:78::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.38.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 192 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'

/etc/yum.repos.d/jenkins.repo                              100%[==========================================================

2025-12-23 09:05:03 (12.1 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [192/192]

[ec2-user@ip-172-31-65-24 ~]$ sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key
[ec2-user@ip-172-31-65-24 ~]$ sudo yum update
Jenkins-stable
Jenkins-stable
Importing GPG key 0xEF5975CA:
 Userid     : "Jenkins Project <jenkinsci-board@googlegroups.com>"
 Fingerprint: 6366 7EE7 4BBA 1F0A 08A6 9872 5BA3 1D57 EF59 75CA
 From       : https://pkg.jenkins.io/redhat-stable/repodata/repomd.xml.key
Is this ok [y/N]: y
Jenkins-stable
Dependencies resolved.
Nothing to do.
Complete!
```

Add required dependencies for the jenkins package:

===

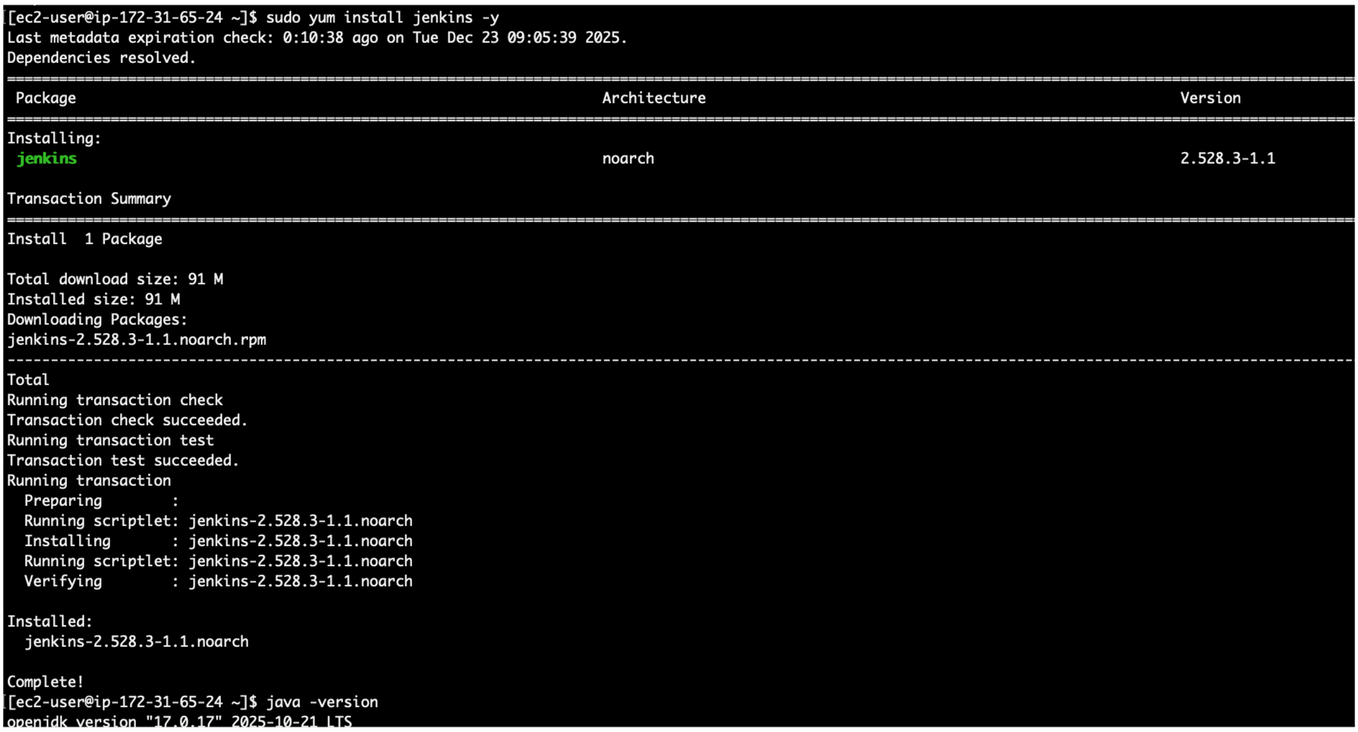
sudo yum install java-17-amazon-corretto -y

```
[ec2-user@ip-172-31-65-24 ~]$ sudo yum install java-17-amazon-corretto -y
Last metadata expiration check: 0:01:06 ago on Tue Dec 23 09:05:39 2025.
Dependencies resolved.
==============================================================================
 Package                                                 Architecture
==============================================================================
Installing:
 java-17-amazon-corretto                                 x86_64
Installing dependencies:
 alsa-lib                                                x86_64
 cairo                                                   x86_64
 dejavu-sans-fonts                                       noarch
 dejavu-sans-mono-fonts                                  noarch
 dejavu-serif-fonts                                      noarch
 fontconfig                                              x86_64
 fonts-filesystem                                        noarch
 freetype                                                x86_64
 giflib                                                  x86_64
 google-noto-fonts-common                                noarch
 google-noto-sans-vf-fonts                               noarch
 graphite2                                               x86_64
 harfbuzz                                                x86_64
```

Install Jenkins:

================

sudo yum install jenkins



```
[ec2-user@ip-172-31-65-24 ~]$ sudo yum install jenkins -y
Last metadata expiration check: 0:10:38 ago on Tue Dec 23 09:05:39 2025.
Dependencies resolved.
================================================================================
 Package                    Architecture                    Version
================================================================================
Installing:
 jenkins                    noarch                          2.528.3-1.1

Transaction Summary
================================================================================
Install  1 Package

Total download size: 91 M
Installed size: 91 M
Downloading Packages:
jenkins-2.528.3-1.1.noarch.rpm
--------------------------------------------------------------------------------
Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :
  Running scriptlet: jenkins-2.528.3-1.1.noarch
  Installing       : jenkins-2.528.3-1.1.noarch
  Running scriptlet: jenkins-2.528.3-1.1.noarch
  Verifying        : jenkins-2.528.3-1.1.noarch

Installed:
  jenkins-2.528.3-1.1.noarch

Complete!
[ec2-user@ip-172-31-65-24 ~]$ java -version
openjdk version "17.0.17" 2025-10-21 LTS
```

Step 2: Change Jenkins Port to 8081

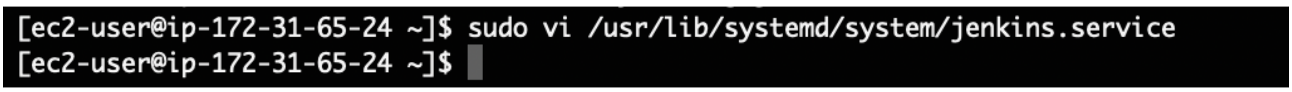
Step-by-Step: Change Jenkins Port to 8081

—> Open Jenkins service file

```
sudo vi /usr/lib/systemd/system/jenkins.service
```

—> Find this line (very important)

```
Environment="JENKINS_PORT=8080"
```



```
[ec2-user@ip-172-31-65-24 ~]$ sudo vi /usr/lib/systemd/system/jenkins.service
[ec2-user@ip-172-31-65-24 ~]$
```

Change **8080 → 8081**

```
# Arguments for the Jenkins JVM
Environment="JAVA_OPTS=-Djava.awt.headless=true"

# Unix Domain Socket to listen on for local HTTP requests. Default is disabled.
#Environment="JENKINS_UNIX_DOMAIN_PATH=/run/jenkins/jenkins.socket"

# IP address to listen on for HTTP requests.
# The default is to listen on all interfaces (0.0.0.0).
#Environment="JENKINS_LISTEN_ADDRESS="

# Port to listen on for HTTP requests. Set to -1 to disable.
# To be able to listen on privileged ports (port numbers less than 1024),
# add the CAP_NET_BIND_SERVICE capability to the AmbientCapabilities
# directive below.
Environment="JENKINS_PORT=8081"

# IP address to listen on for HTTPS requests. Default is disabled.
#Environment="JENKINS_HTTPS_LISTEN_ADDRESS="

# Port to listen on for HTTPS requests. Default is disabled.
# To be able to listen on privileged ports (port numbers less than 1024),
-- INSERT --
```

Step 3: Start Jenkins

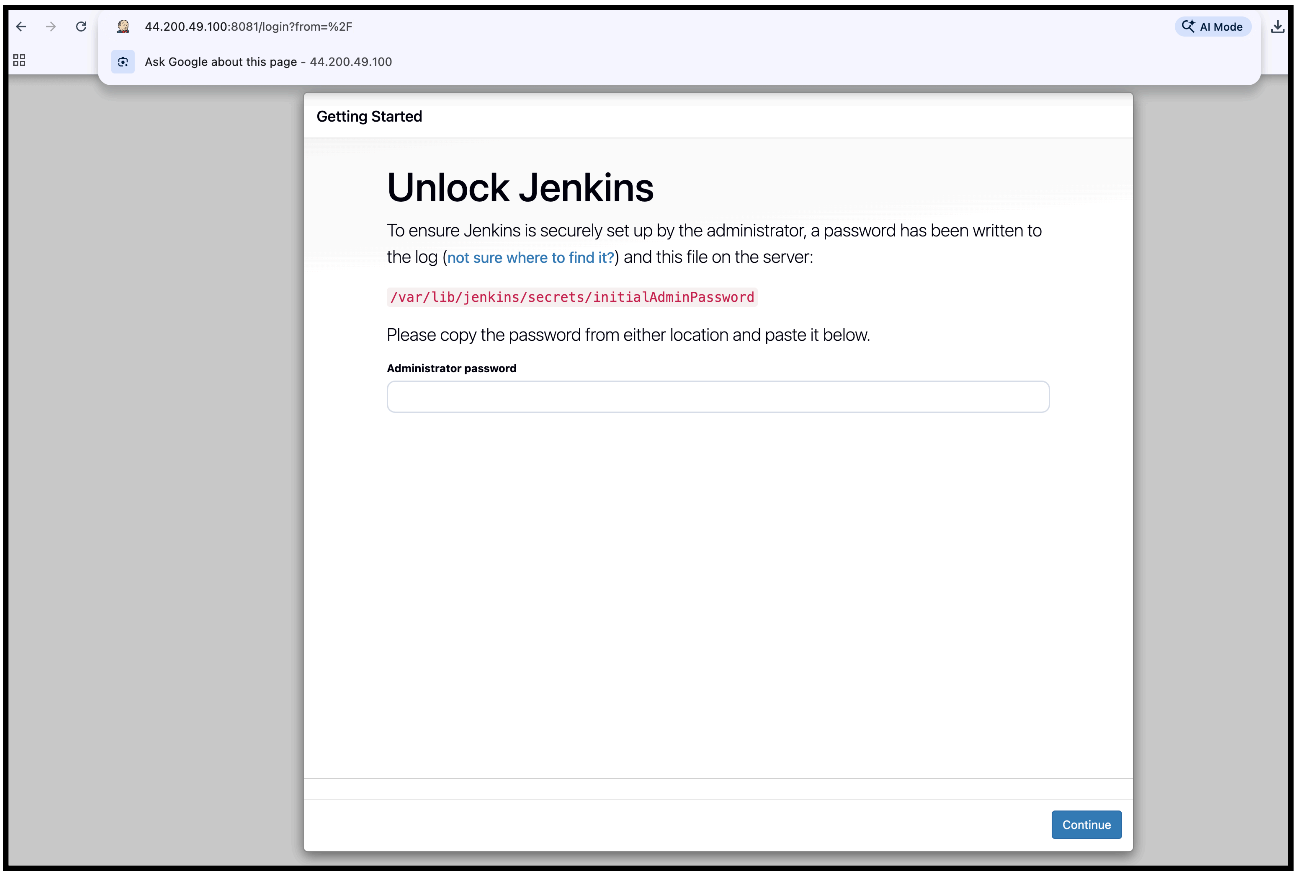
```
sudo systemctl daemon-reload
sudo systemctl start jenkins
sudo systemctl enable jenkins
Sudo systemctl status jenkins
```

```
[ec2-user@ip-172-31-65-24 ~]$ sudo systemctl daemon-reload
[ec2-user@ip-172-31-65-24 ~]$ sudo systemctl start jenkins
[ec2-user@ip-172-31-65-24 ~]$ sudo systemctl enable jenkins

[ec2-user@ip-172-31-65-24 ~]$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
     Active: active (running) since Tue 2025-12-23 09:25:44 UTC; 44s ago
   Main PID: 26357 (java)
      Tasks: 43 (limit: 1067)
     Memory: 398.3M
        CPU: 18.082s
     CGroup: /system.slice/jenkins.service
             └─26357 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/

Dec 23 09:25:39 ip-172-31-65-24.ec2.internal jenkins[26357]: [LF]>
Dec 23 09:25:39 ip-172-31-65-24.ec2.internal jenkins[26357]: [LF]> *************************************************
Dec 23 09:25:39 ip-172-31-65-24.ec2.internal jenkins[26357]: [LF]> *************************************************
Dec 23 09:25:39 ip-172-31-65-24.ec2.internal jenkins[26357]: [LF]> *************************************************
Dec 23 09:25:44 ip-172-31-65-24.ec2.internal jenkins[26357]: 2025-12-23 09:25:44.237+0000 [id=30]        INFO
Dec 23 09:25:44 ip-172-31-65-24.ec2.internal jenkins[26357]: 2025-12-23 09:25:44.253+0000 [id=23]        INFO
Dec 23 09:25:44 ip-172-31-65-24.ec2.internal systemd[1]: Started jenkins.service - Jenkins Continuous Integration
Dec 23 09:25:44 ip-172-31-65-24.ec2.internal jenkins[26357]: 2025-12-23 09:25:44.378+0000 [id=49]        INFO
Dec 23 09:25:44 ip-172-31-65-24.ec2.internal jenkins[26357]: 2025-12-23 09:25:44.379+0000 [id=49]        INFO
Dec 23 09:25:49 ip-172-31-65-24.ec2.internal jenkins[26357]: 2025-12-23 09:25:49.307+0000 [id=69]        WARNING
[ec2-user@ip-172-31-65-24 ~]$
```

Access Jenkins



—> jenkins is running in port 8081

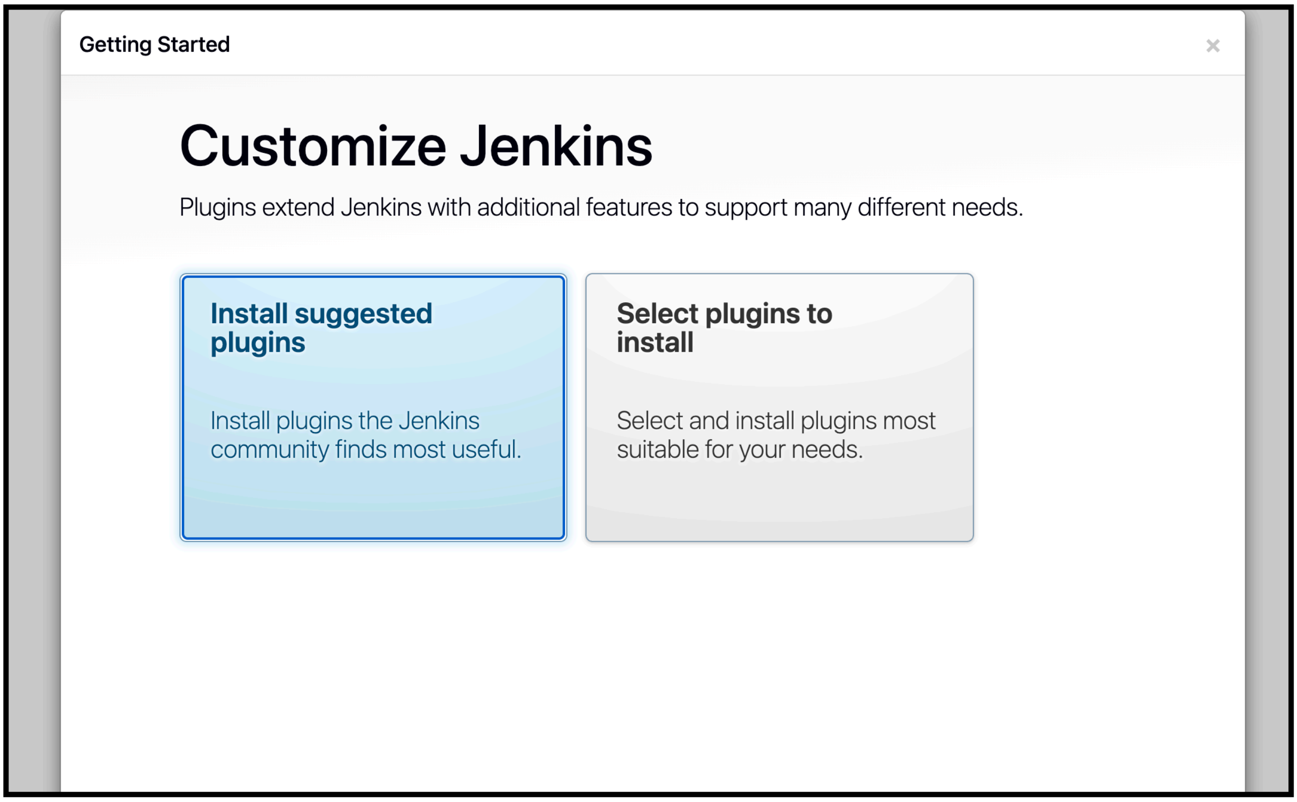
Now we have to enter the administration password

sudo cat /var/lib/jenkins/secrets/initialAdminPassword

```
[ec2-user@ip-172-31-65-24 ~]$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
4b8144211f2a462c884717d481c8fbf9
[ec2-user@ip-172-31-65-24 ~]$
```

Copy the password and login

Click: Install suggested plugins



After Plugins Installation (IMPORTANT)

You will see **Create First Admin User** screen.

Fill like this (example):

- Username: `admin`
- Password: (your choice)
- Confirm password
- Full name: Admin
- Email: optional

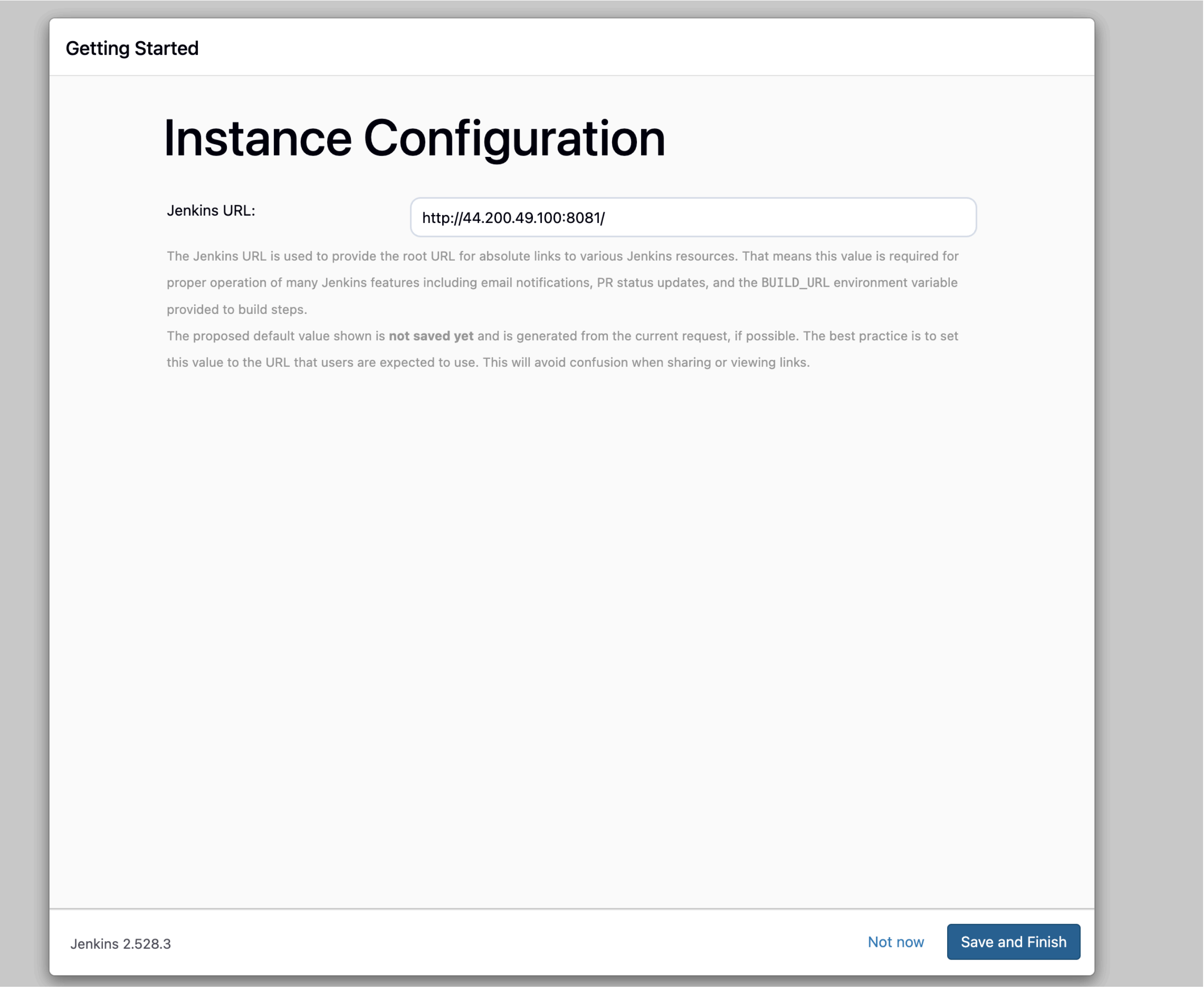
Click **Save and Continue**

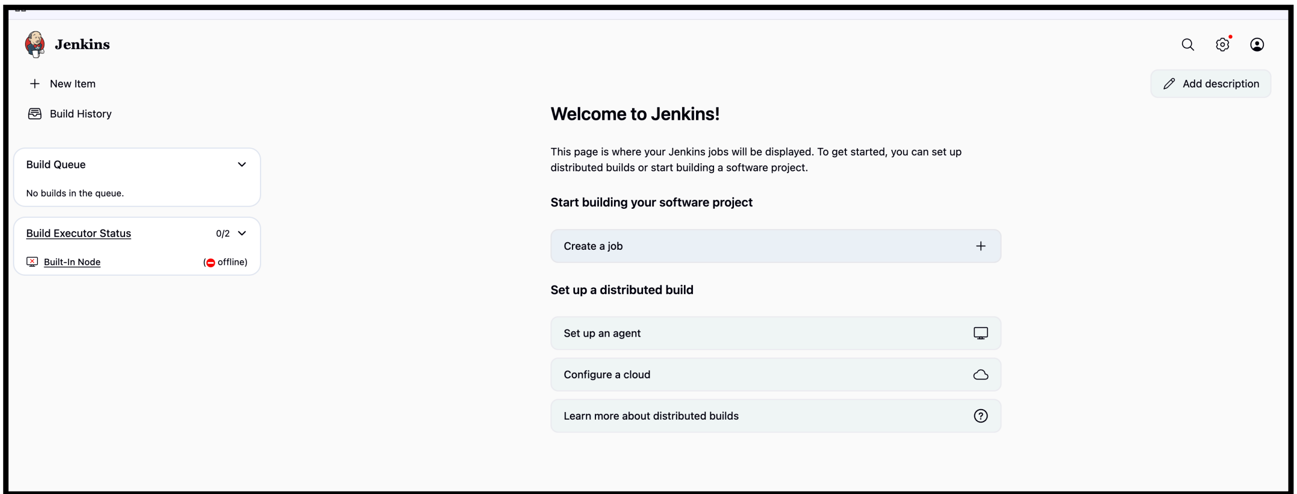
Jenkins URL

You’ll see:

```
http://<EC2-PUBLIC-IP>:8081/
```

Click **Save and Finish → Start using Jenkins**

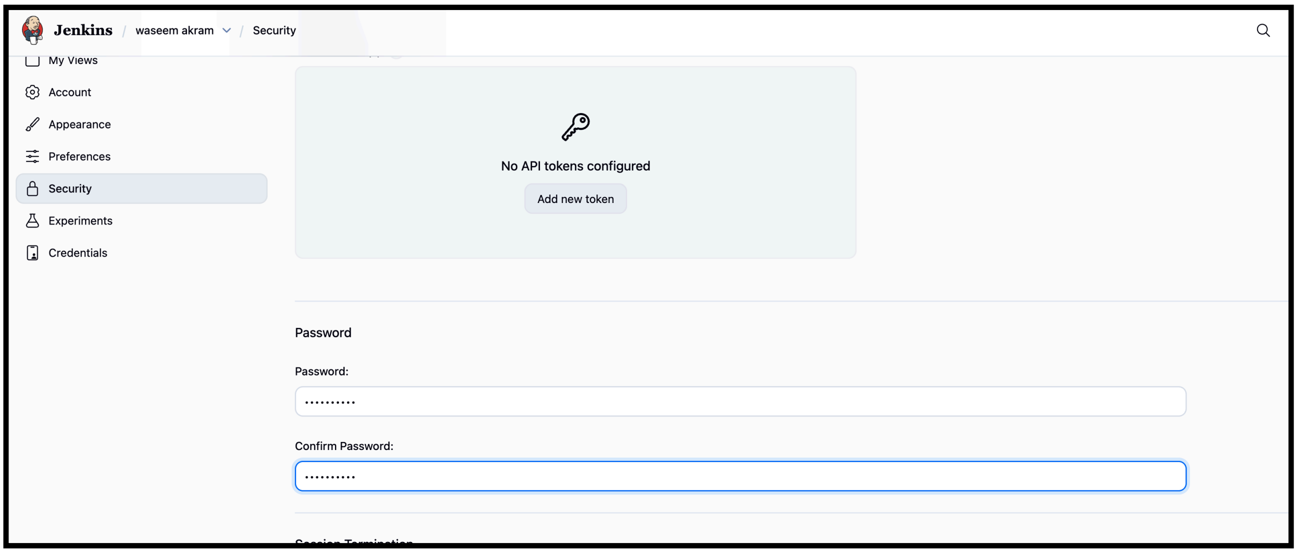




—> now we are at now on the Jenkins Dashboard. Installation is successfully completed

2. Secure Jenkins Server

—> change the password in security and remove the



—> and removed the initial admin password

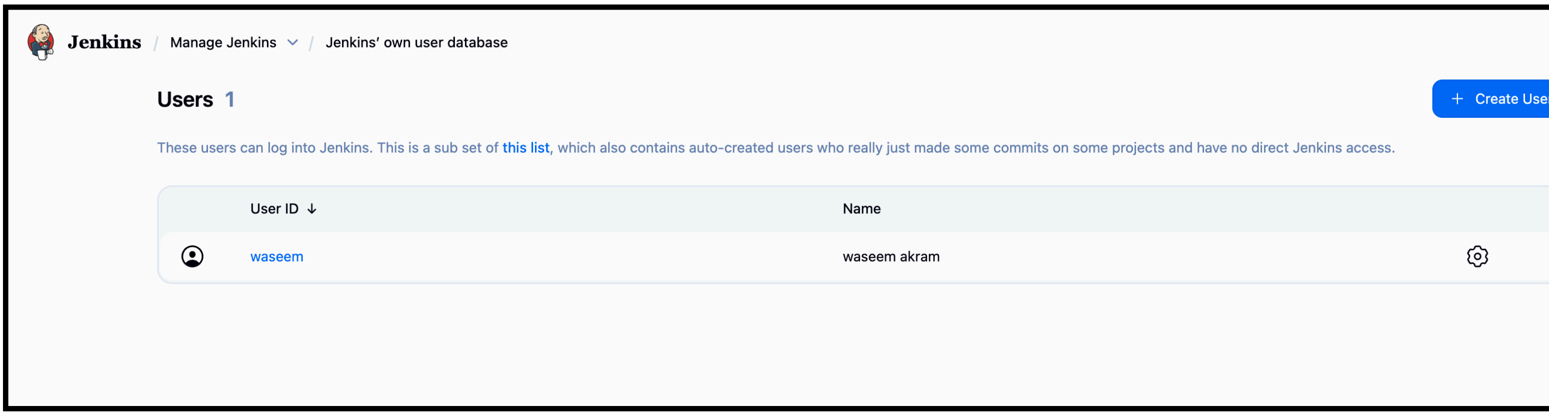
```
[ec2-user@ip-172-31-65-24 ~]$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
cat: /var/lib/jenkins/secrets/initialAdminPassword: No such file or directory
[ec2-user@ip-172-31-65-24 ~]$
```

—-> this will make the Jenkins login secure if any hacker gets the initialadminpassword but he will not be able to login into our Jenkins server because we changed the password and removed the initialadminpassword

3.Create users called DevOps, Testing in Jenkins with Limited access.

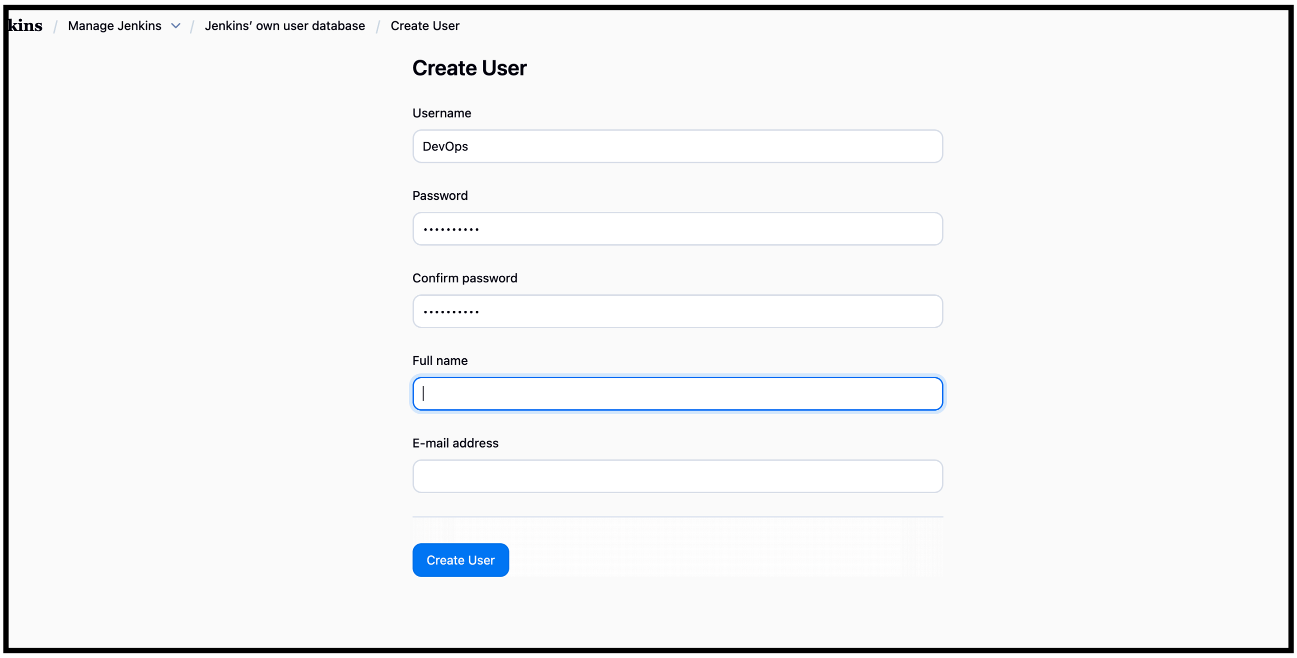
Create Users

Go to **Manage Jenkins → Manage Users**



Create DevOps User

1. Click **Create User**
2. Enter the following details:
    - **Username:** DevOps
    - **Password:** set password
    - **Confirm Password:** same password
    - **Full Name:** DevOps User
    - **Email:** optional
3. Click **Create User**



Create Testing User

1. Click **Create User**
2. Enter the following details:
    - **Username:** Testing
    - **Password:** set password

    - **Confirm Password:** same password
    - **Full Name:** Testing User
    - **Email:** optional
3. Click **Create User**

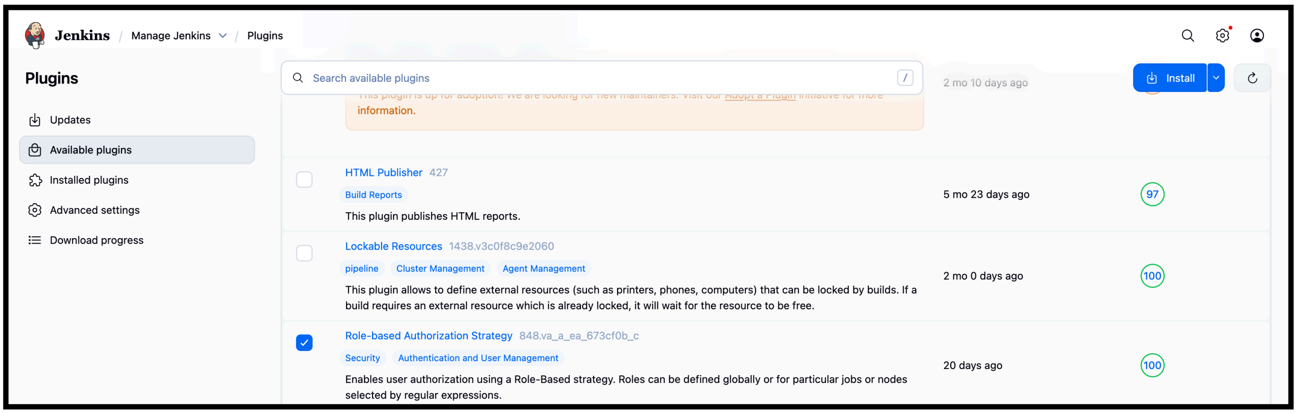
Verify Users

Ensure the following users exist:

- Admin user (created during setup)
- DevOps
- Testing

—> now for giving limited access we have to install a plugins

—> and select the role-based authorisation strategy



Then go to security for authanticater tab.

Then gave a permissons to the user.

—> and save

Final Result

- Jenkins admin user exists
- DevOps and Testing users created successfully
- Users can log in to Jenkins

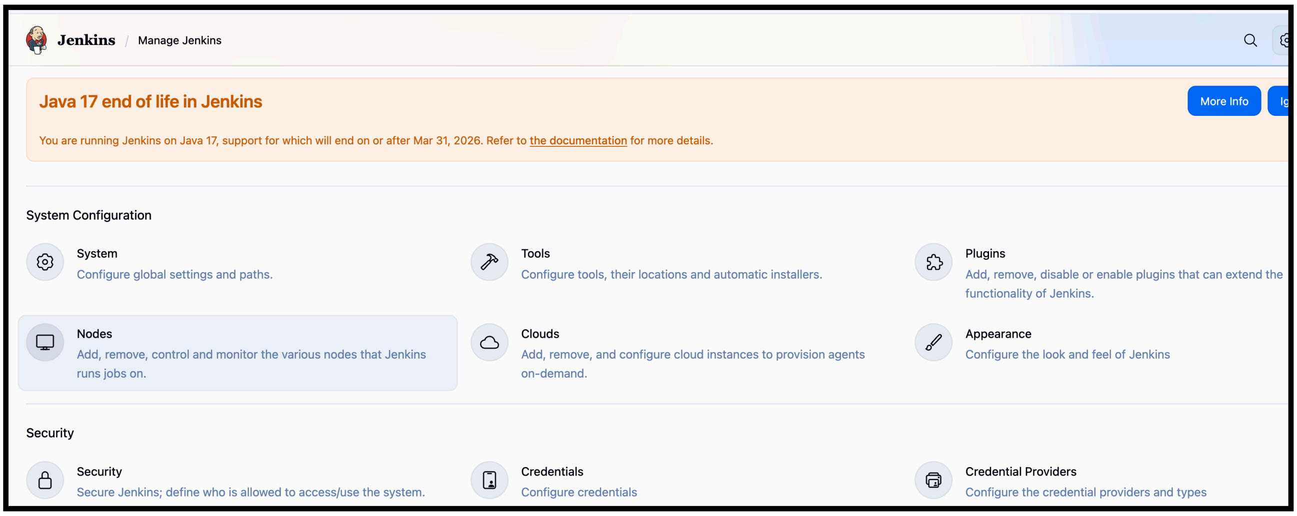
4. Configure Labels & Restrict Jobs to Run on Label Only

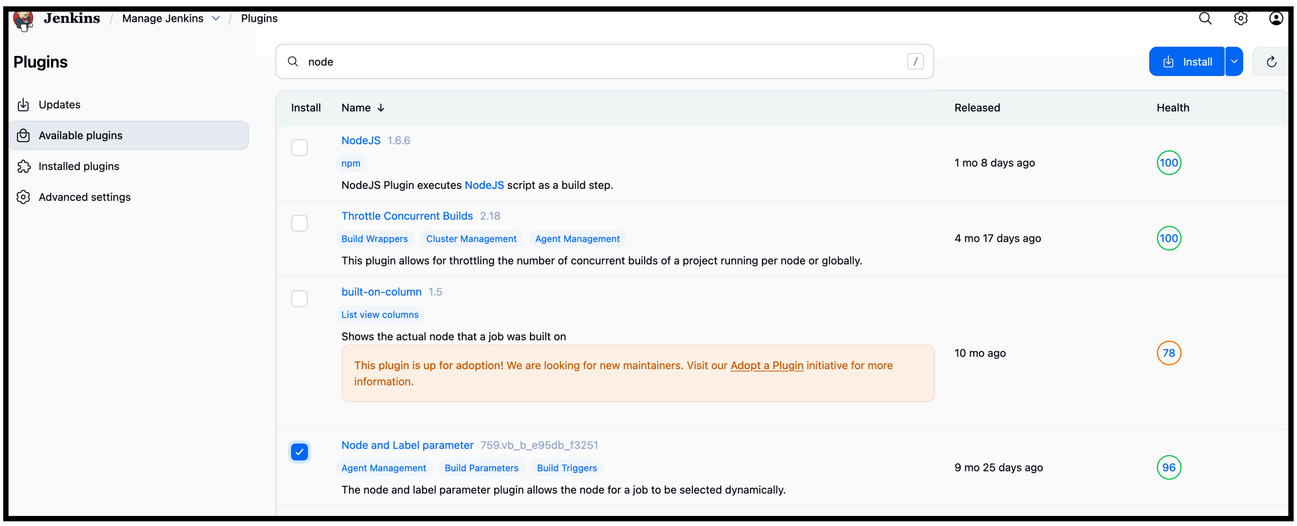
—> Here the explanation about label and restrict jobs.

Go to manage Jenkins first.

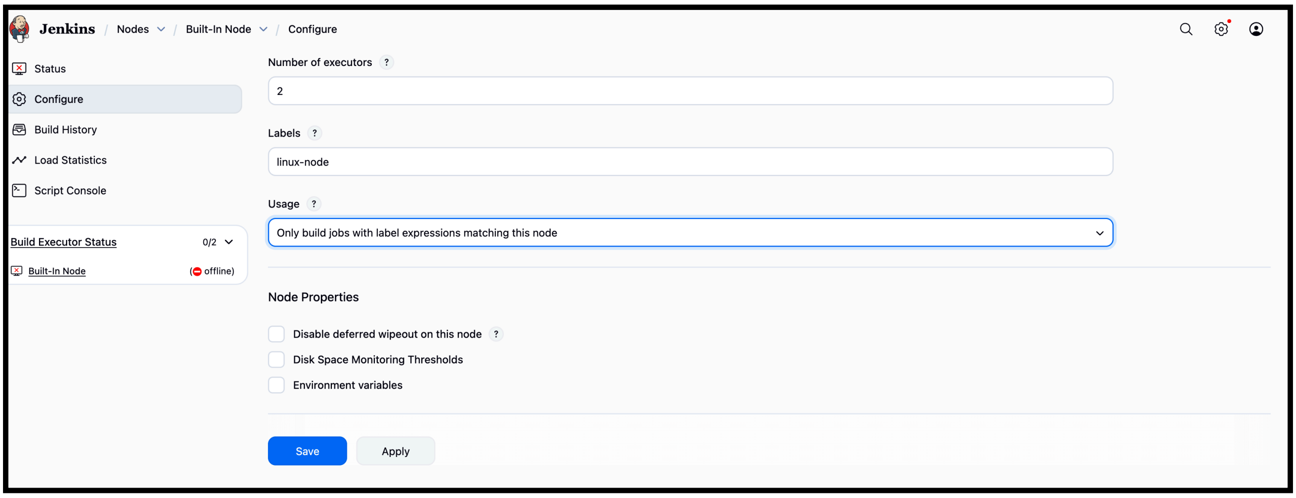
Then go to plugins >> nd available plugins >> node label

parameter >> install.





—> Then in manage Jenkins >> node > check the node. And save

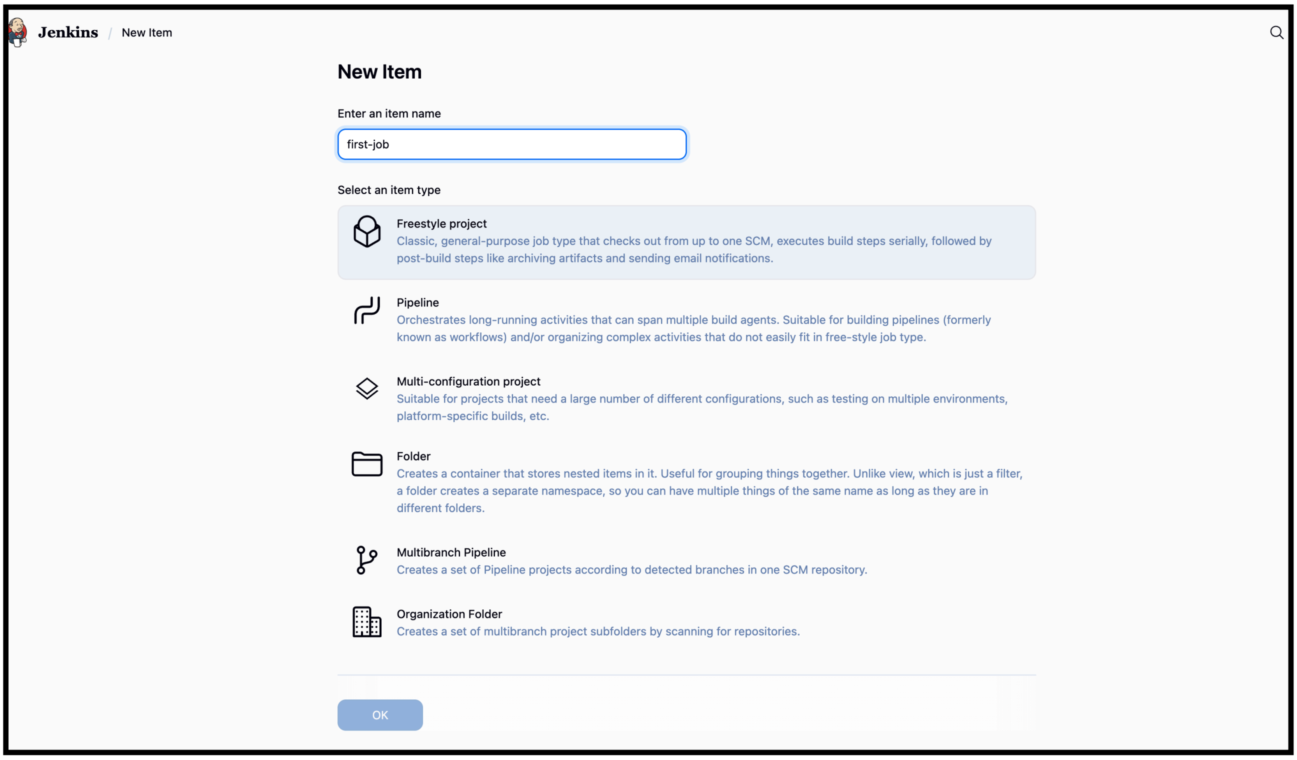


—-> change the free space threshold value to 0.1 gb in node conf

➢ Then go to the dashboard >> create job >> gave a URL its shows you error first.

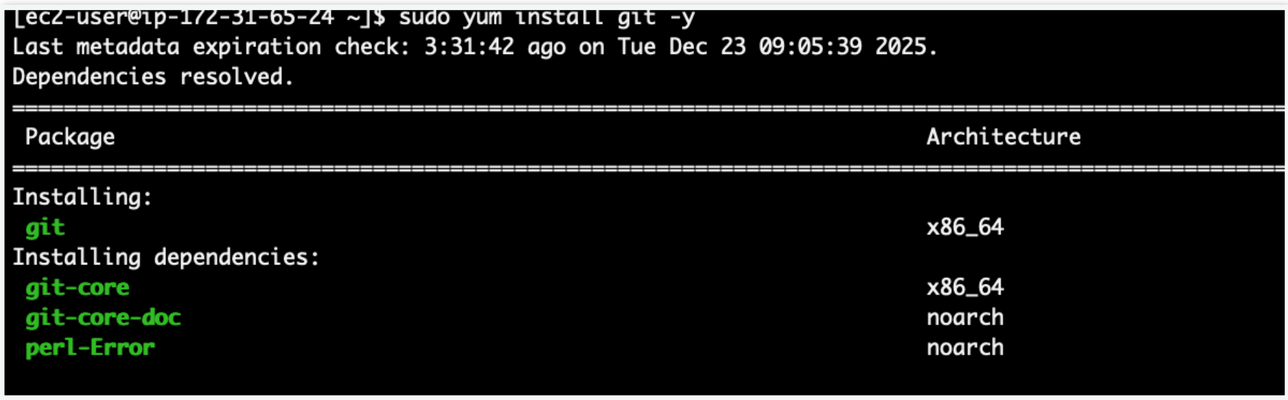
—-> create a job and in general add git url

https://github.com/betawins/Techie_horizon_Login_app.git

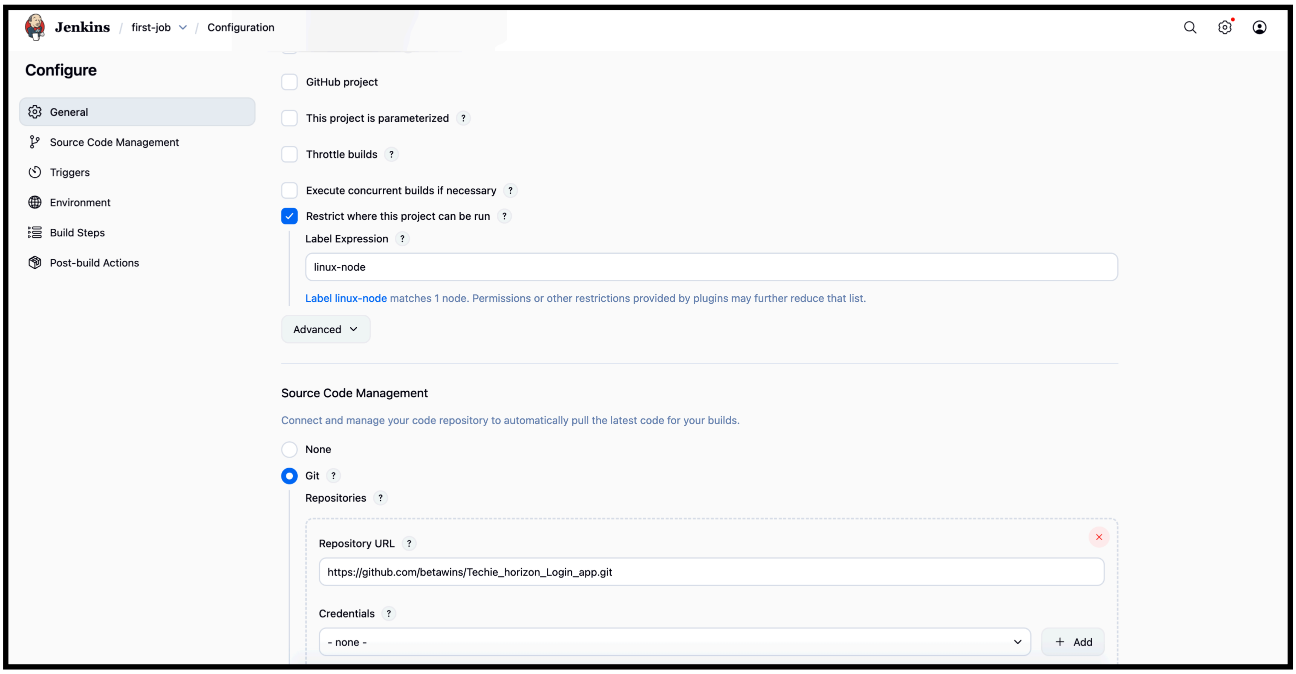


—-> Then connect to the your instance nd check the git version.

Sudo yum install git -y

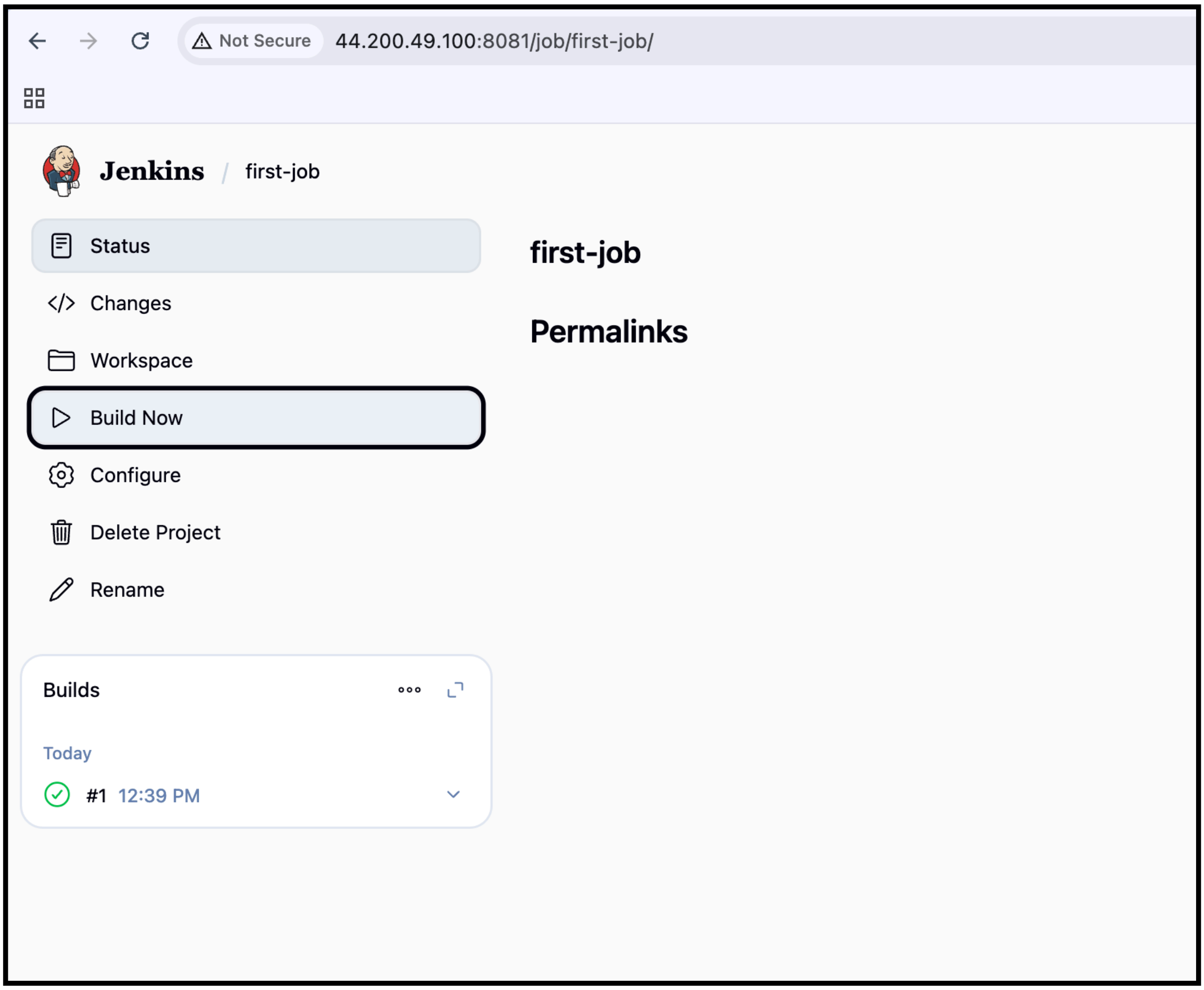


- If its not available then use “yum install git -y”
- Then the error will gone.



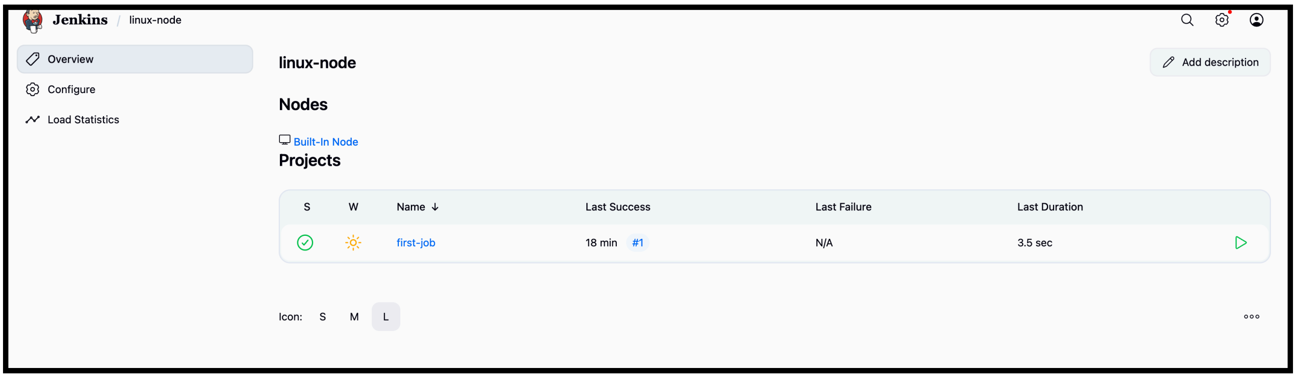
—> and save changes

—> now click build now



—> check console output

—> now check the node labels



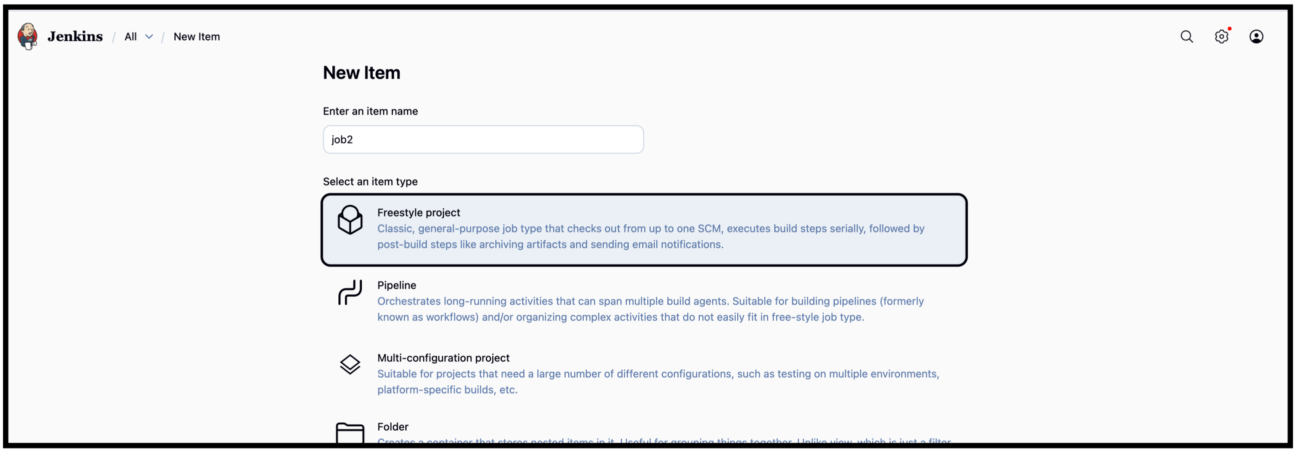
5. Create Three sample jobs using the below URL. https://github.com/betawins/Techie_horizon_Login_app.git

Create Three Sample Jenkins Jobs

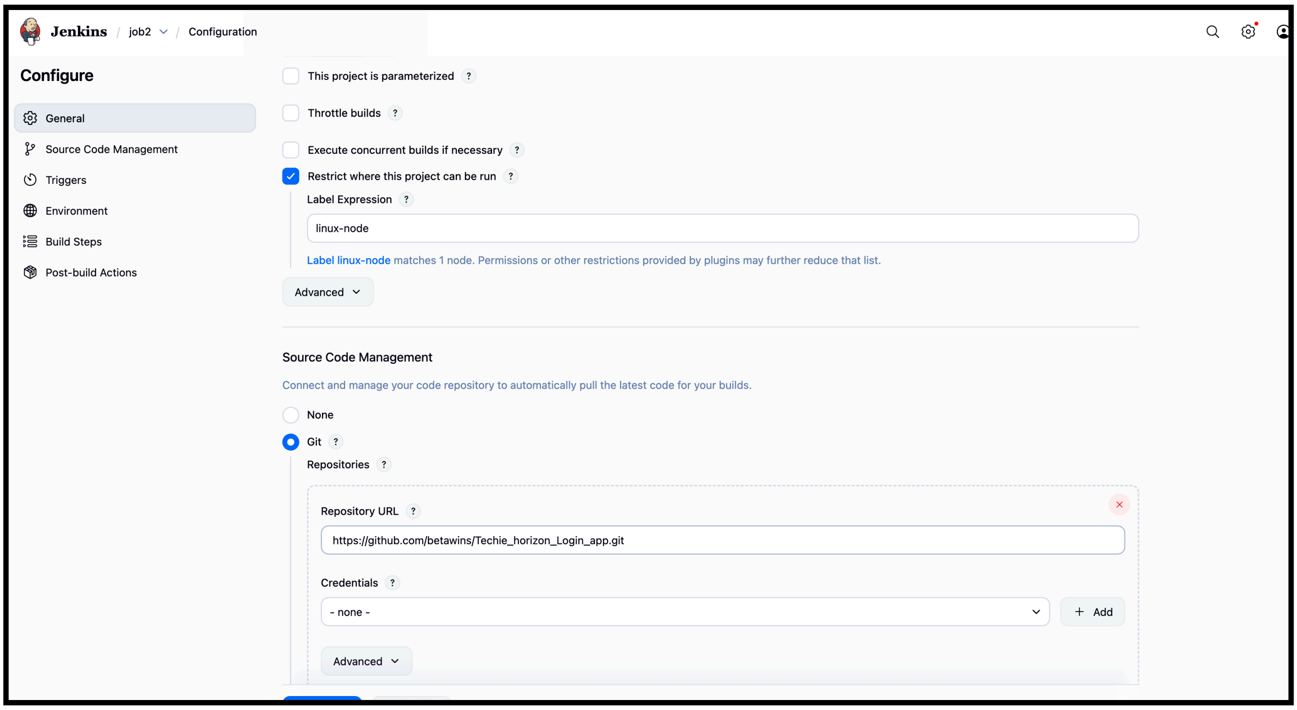
GitHub Repository Used:

```
https://github.com/betawins/Techie_horizon_Login_app.git
```

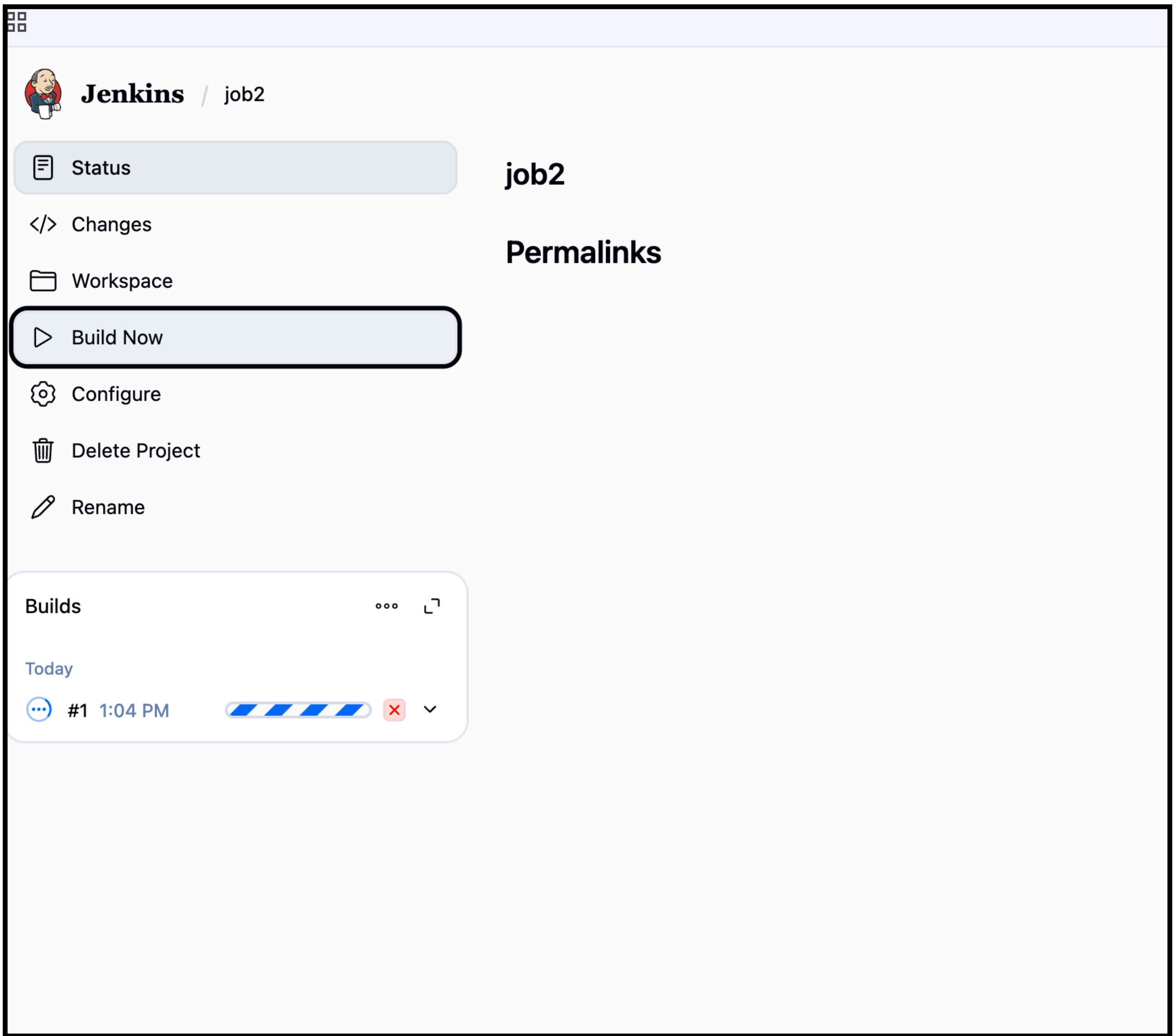
- —>Go to the dashboard of jenkins.
- Create a new item.
- Then gave a name of that.in discription.
- Nd choose git nd copy the repo url.
- And save.
- Do same thing other 2 jobs also.



—> create the item (job)

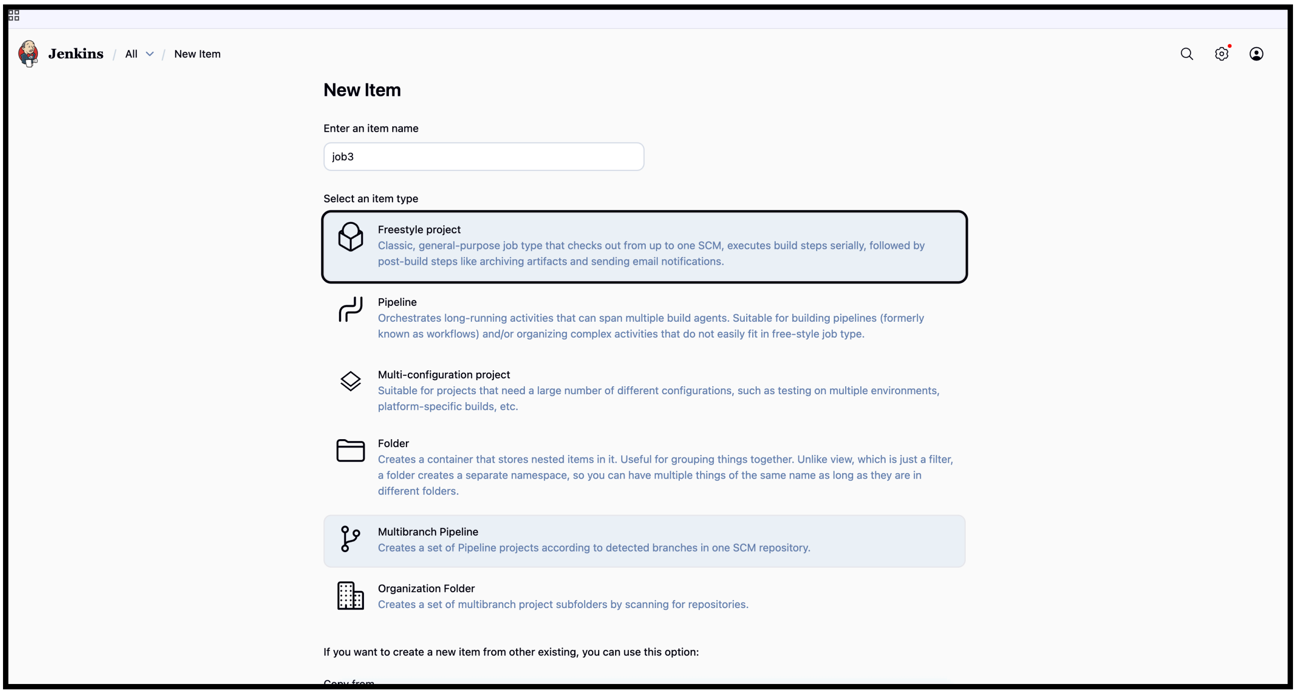


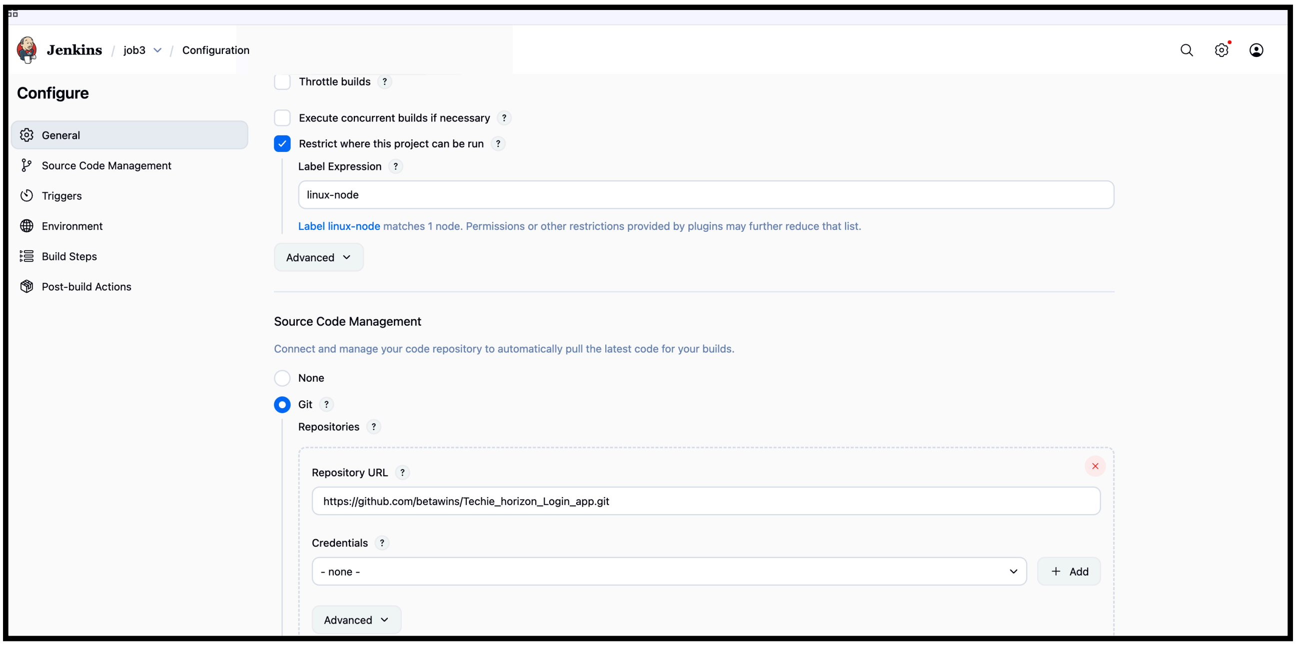
—> now add the label and git url and save

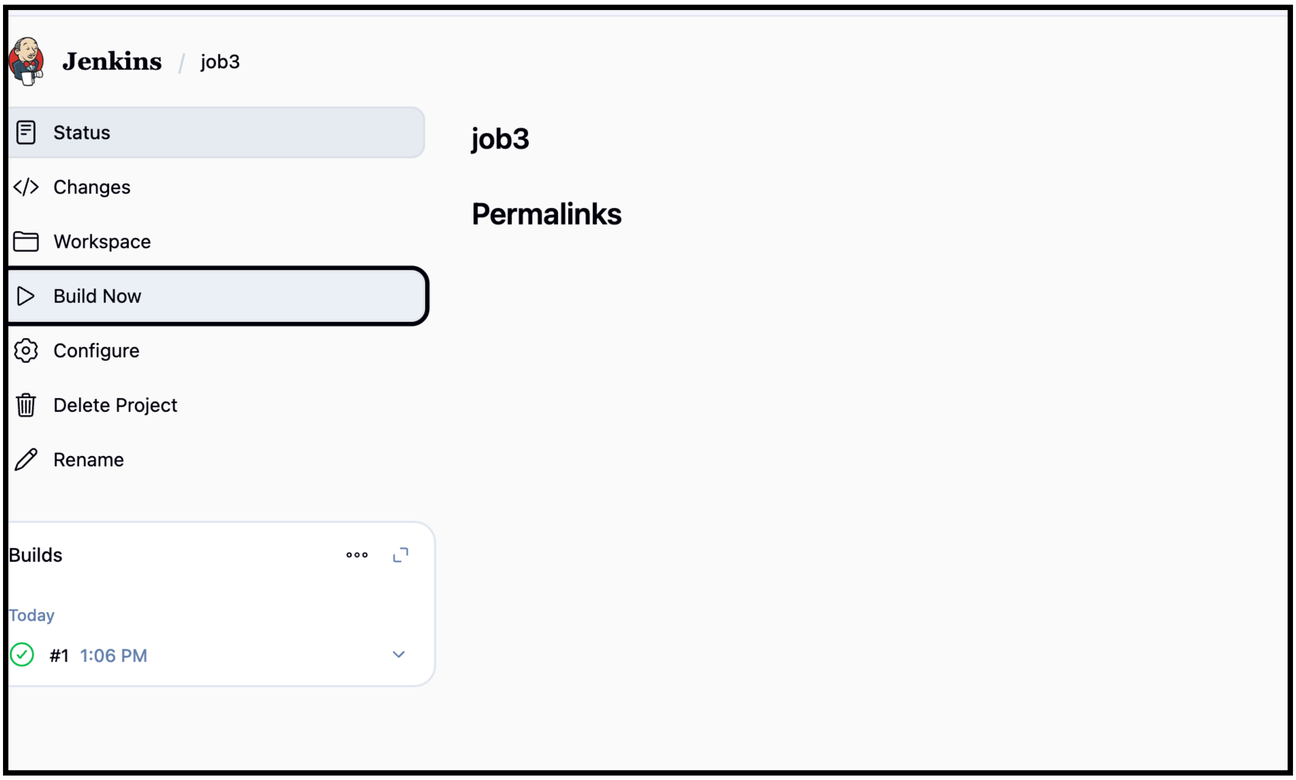


—> click build now

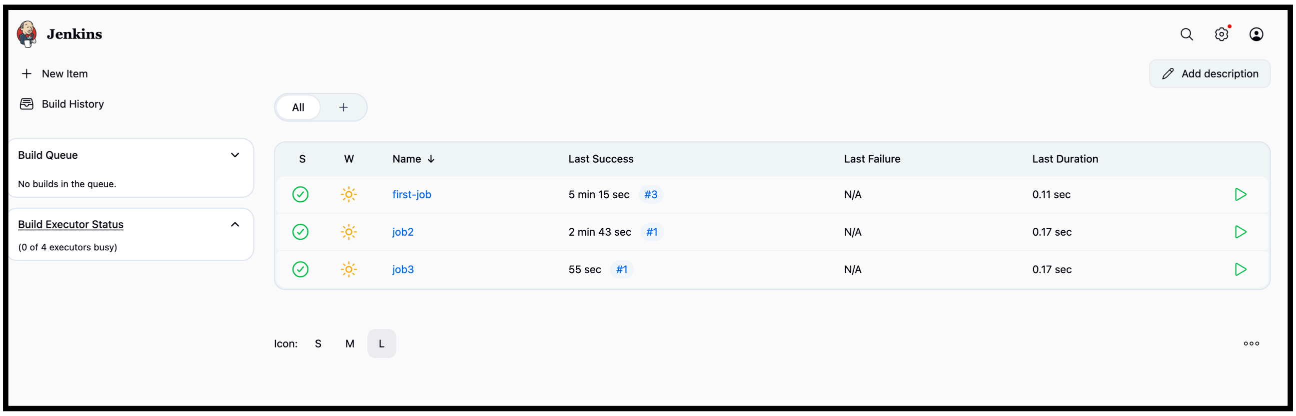
Similar do for other jobs







—-> after create a new job and attach the git url similar to above job also

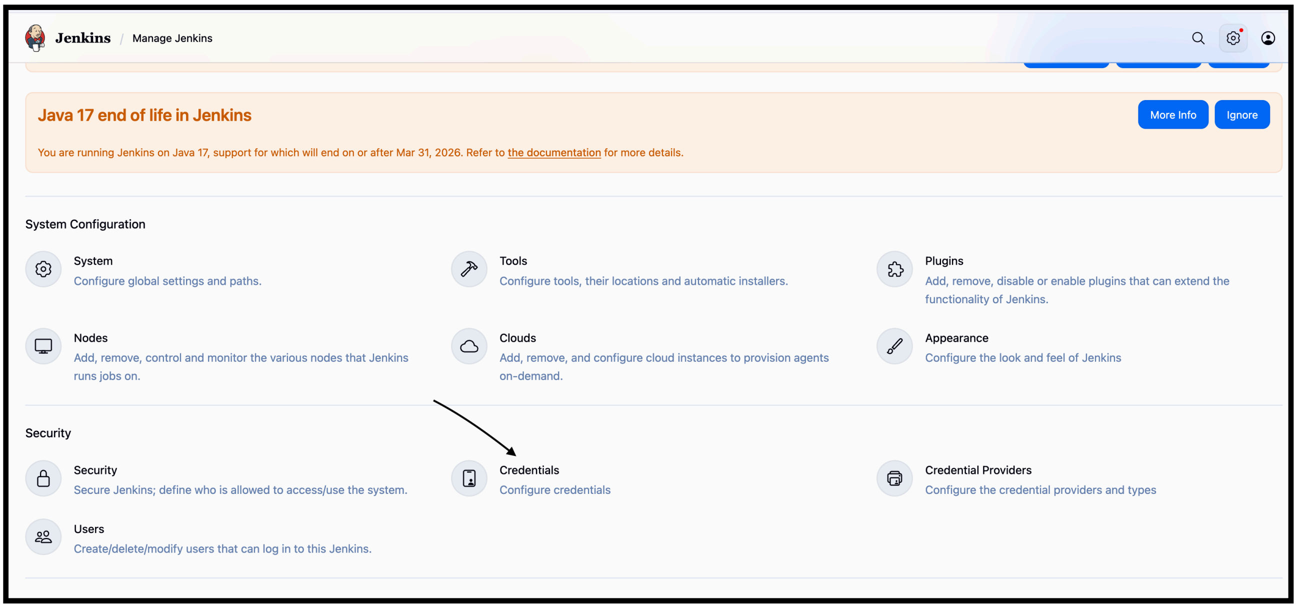


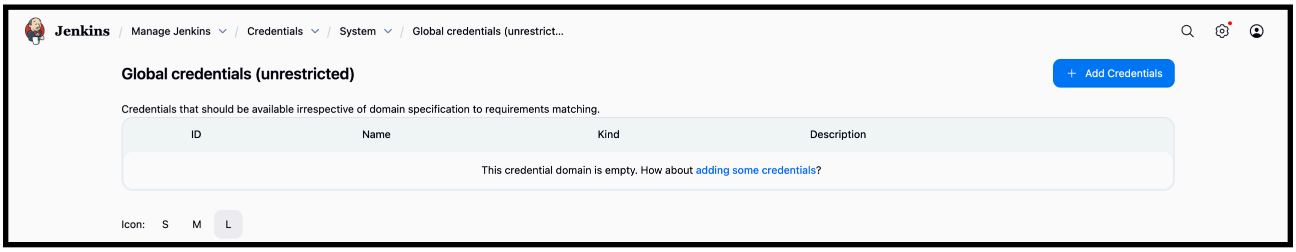
—-> hence three sample jobs created with URL. https://github.com/betawins/Techie_horizon_Login_app.git

6) Create one Jenkins job using git hub Private repository.

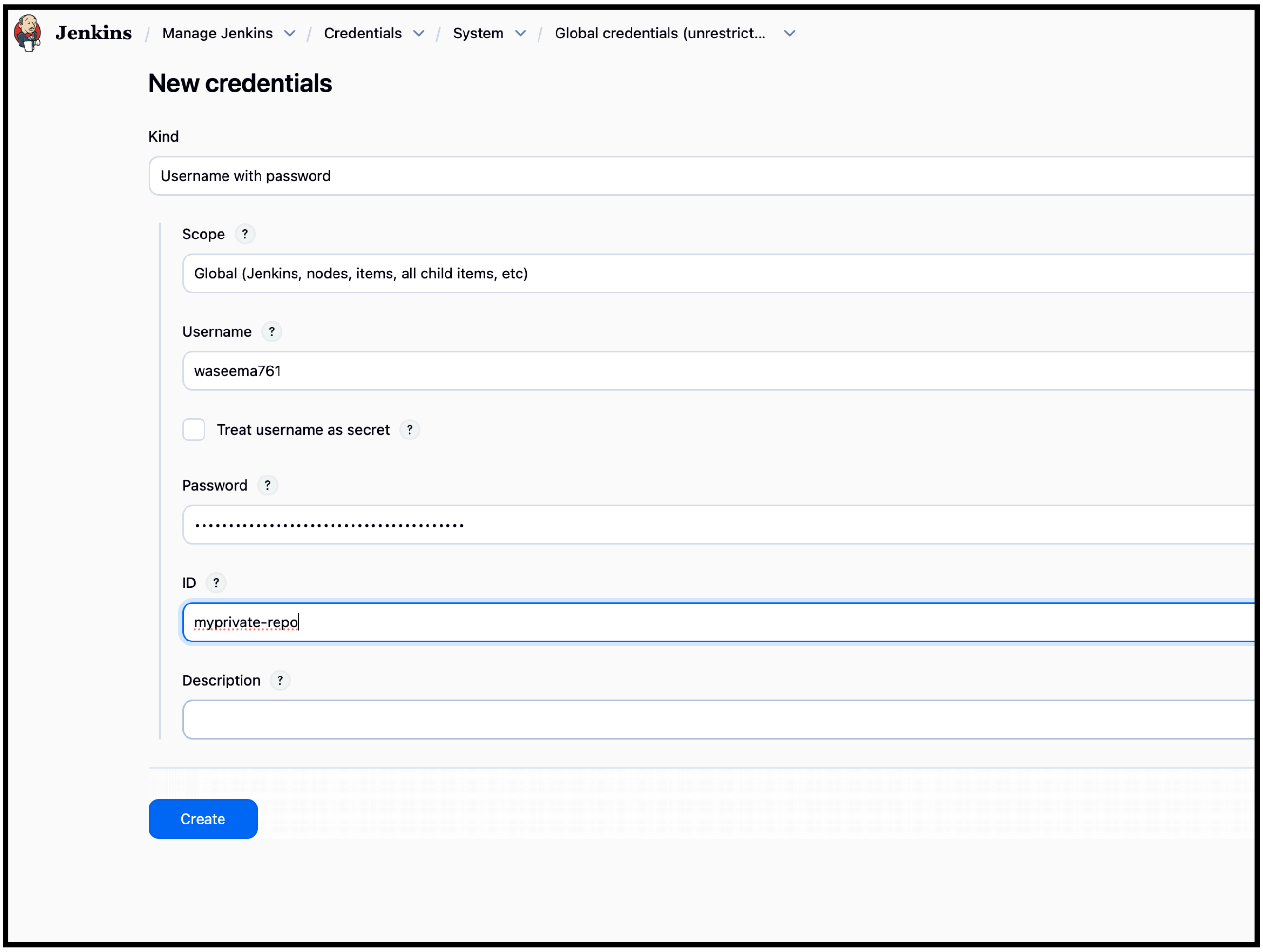
—>Go to manage jenkins.

- ➢ Then use credentials.
- ➢ Then system and create credentials
- ➢ And save it

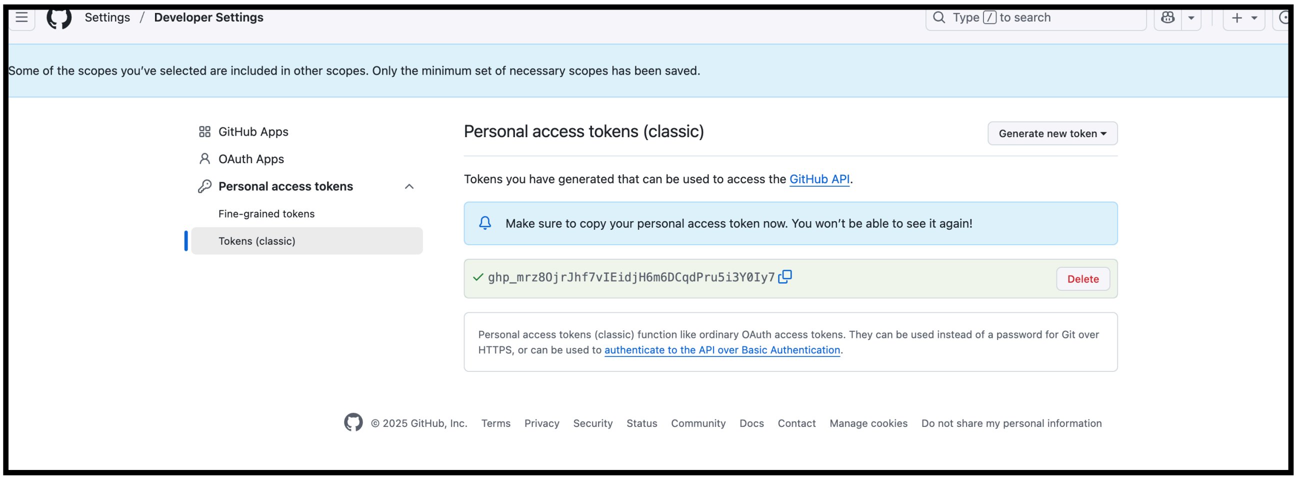




—-> fill the details of the private repo from user

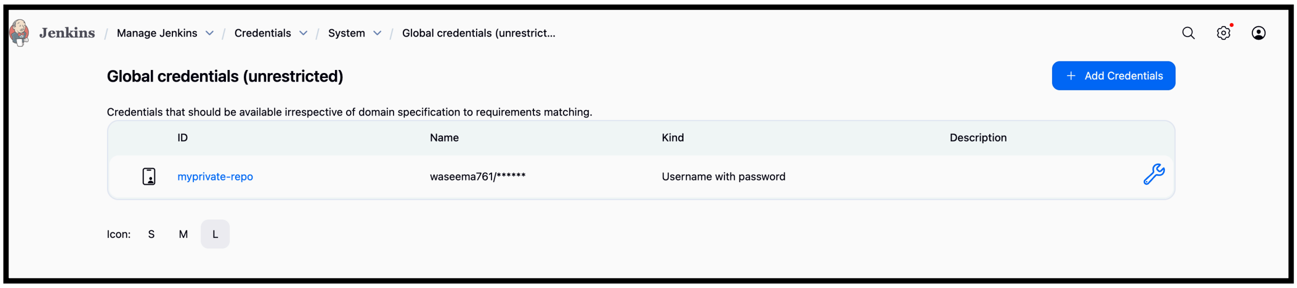


—> go to your GitHub and create a token and copy it and paste on



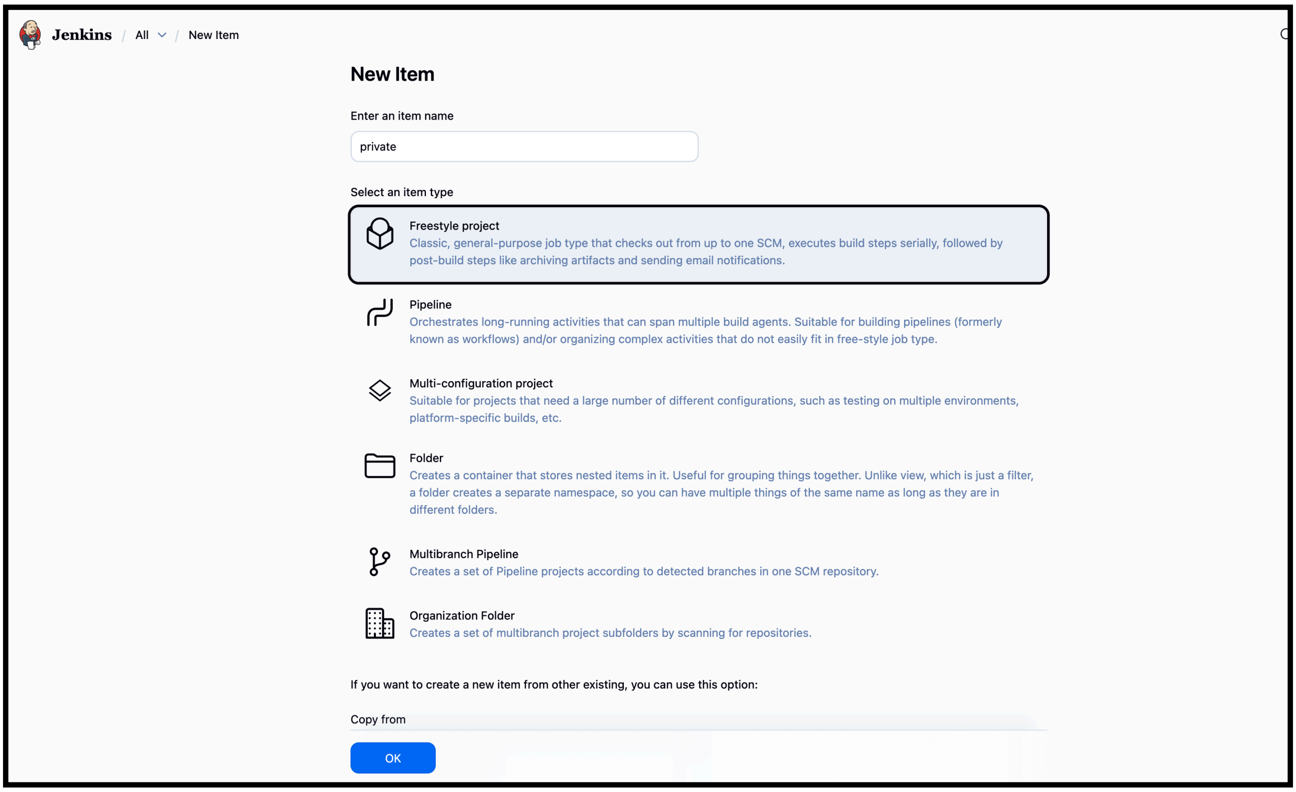
above password in credentials

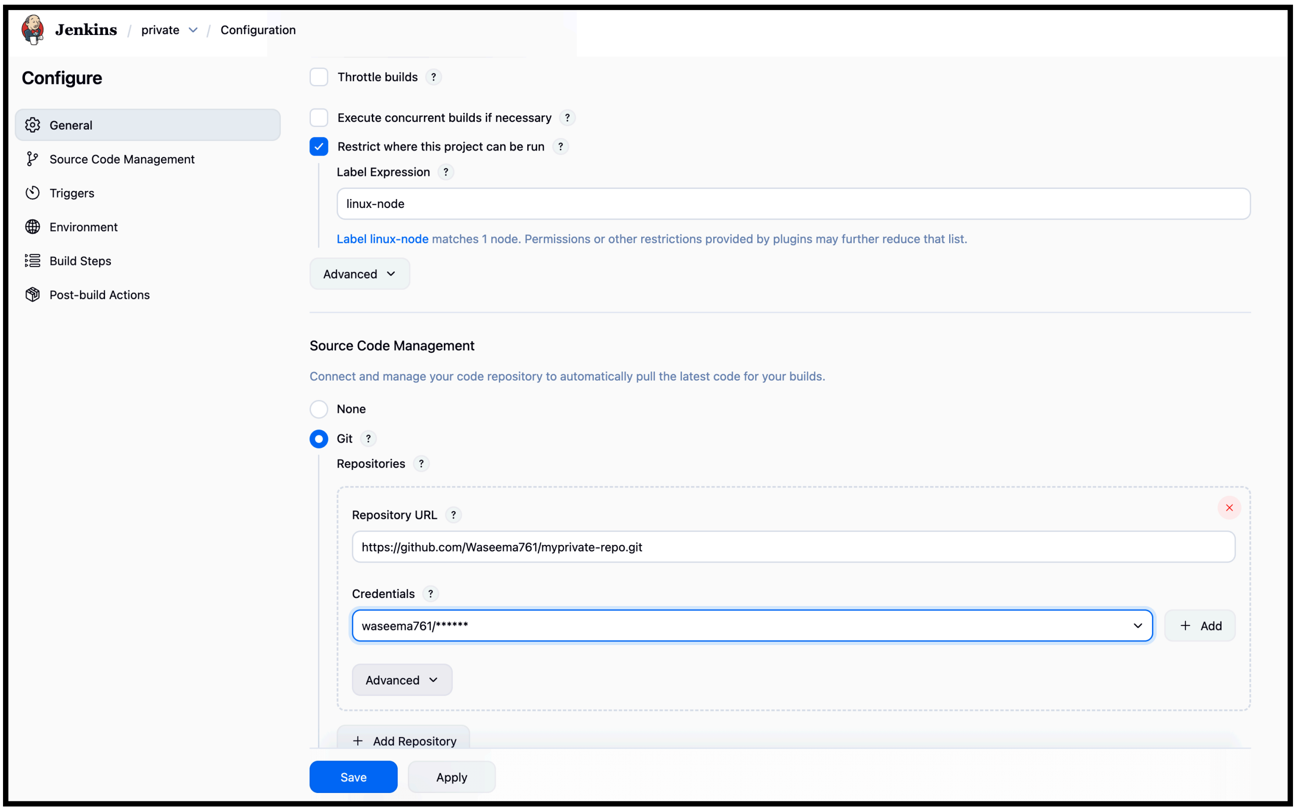
—> now after creating the credentials



—>Go to jenkins dash board.

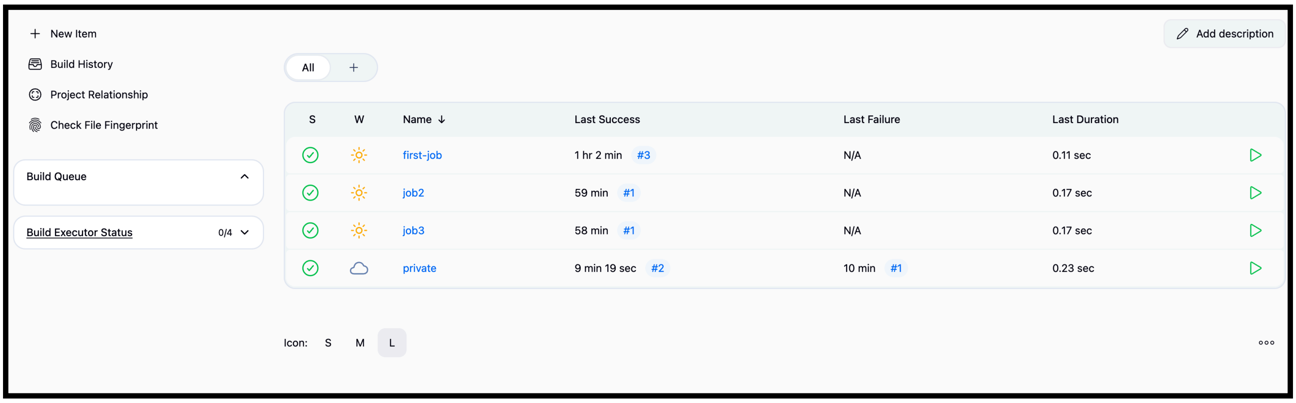
- ➢ Then create a job.
- ➢ Use git.
- ➢ Give your private repo http address for that.





—> inside git add the private repo url and add the credentials which we created

➢ Then go to build >> excute build steps -> console output



—> hence Create one Jenkins job using git hub Private repository.